

Even though Nepal is a small country in terms of Geographical area, it is highly rich in natural resources. Some of the major natural resources of Nepal are

- Water/Rivers
- Forest
- Conservation Areas
- Hunting Wildlife Reserve
- Minerals and their ores
- Lakes, ponds, Daha, Kunda, water falls, glaciers
- National Parks
- Wildlife Reserve
- Soil or Land, wetlands

Water resources

- Nepal is regarded the second richest country in terms of water resources after Brazil.
- Nepal occupies 2.27% of the total water resource of the world whereas Brazil occupies 5.67%.
- The bodies of water here are regarded as the key strategic natural resources with the potential to act as the catalyst for the all-round development and economic growth of the country.
- There are many sources of water in Nepal, which are
 - ❖ Rivers
 - ❖ Lakes, Ponds, Kunda, Daha
 - ❖ Glaciers
 - ❖ Water falls etc

Rivers

- Nepal is a landlocked country which means Nepal has no direct connection to any sea route. But there are lots of rivers that flow from the Himalayas. When snow melts in the Himalayas, the glaciers and rivers are formed. The rivers flow through the mountain regions to the Terai.
- Area occupied by rivers and rivulets is 3.9% of the total area of Nepal.
- There are about 6000 rivers in Nepal with a total drainage area of 194,471 sq. km. Out of this, 74 % lies within the country.
- The total length of all rivers and rivulets within the country is estimated to be 45,000km.
- The rivers of Nepal finally end in Ganga River in India and Ganga ends in Indian Ocean.
- It is estimated that the rivers of Nepal can produce 83000 megawatt (8 crore and 30 lacs kilowatt) of hydroelectricity. This data was presented by Dr. Hariman Shrestha as a part of thesis dissertation.
- Out of this estimated potential, 114 projects with a combined capacity of 42,000 MW have been identified as economically and technically feasible.
- **The currency that is obtained by exporting/selling hydroelectricity is called White Dollar.**

Current Affairs:

- The irrigation facility in the current fiscal year has reached to 15 lakhs nine thousand four hundred and twenty seven hectare until Asar 2078.
- Currently 16 irrigation projects are in operation under Federal government.

Classification of rivers of Nepal

- **Based on the place of origin, rivers of Nepal are classified into 3 different classes**
 - **First Class rivers/Evergreen rivers:** Rivers originating from the Himalayas
 - ❖ Principal purpose: Generation of Hydro Electricity.
 - ❖ Principal rivers: Sapta-Koshi, Sapta-Gandaki, Sapta-Karnani and Mahakali.
 - **Second Class rivers:** Rivers originating from Mahabharata Range
 - ❖ Principal purpose: Irrigation and least percentage of Hydro electricity generation.
 - ❖ Principal rivers: Mechi, Kankai, Triyuga, Kamala, Banganga, Tinau, Rapti, Mohana, Babai, Jhimruk, Bagmati, Manohara, Hanumante, Bishnumati, Karmonasa etc.

- **Third Class rivers:** Rivers originating from Chure Range
 - ❖ These types of rivers are seasonal rivers.
 - ❖ Principal rivers: Sirsiya, Ratuwa, Sunsari, Tilawe, Jamuni, Hardinath.
 - ❖ These rivers are mostly responsible for floods and calamities.

Rivers and Rivulets of Nepal

First class rivers

S.N	Features	Koshi	Gandaki	Karnali
1	Size	largest	Deepest	longest
2	Length (km)	720	338	507
3	Electricity generation Capacity	22000 Megawatt	21000 Megawatt	32000 Megawatt
5	Largest Tributary	Arun	Kali gandaki	Humla Karnali
6	Longest Tributary	Arun	Marsyangdi	Humla karnali
7	Smallest Tributary	Likhu	Trishuli/Daraudi	Tila

Koshi River

- Named after the name of a Shage "Kaushik".
- The ancient name of Koshi river is Kumari river.
- Tributaries: Tamor, Arun, Dudhkoshi, Likhu, Tamakoshi, Sunkoshi, Indrawati (East to West).
- Second largest sand carrying capacity river in the world.
- Koshi River is called Sorrow of Bihar and Sorrow of Bengal.
- Koshi River in China is called "Sangpo".

Gandaki/ Narayani River:

- Gandaki River in Terai region is called Narayani river.
- Tributaries: Trishuli, Budhigandaki, Daraudi, Marshyangdi, Madi, Seti, Kaligandaki (East to West).
- Kaligandaki River is famous for a precious black stone called "Shaligram".
- Kaligandaki is also called cyclonic (Chakrakar) river of Nepal.

Karnali

- Tributaries: Sano Bheri, Thuli Bheri, Mugu Karnali, Tila, Humla Karnali, Seti, Budhi Ganga (East to West).
- Karnali River is famous for Dolphin named Sonsh, 1st seen in 2038 B.S.

The rivers of Nepal in India are called:

- | | |
|--------------------|---------------------|
| ▪ Mechi: Mahanda | ▪ Koshi: Damodar |
| ▪ Gandaki: Gandak | ▪ Karnali: Ghangara |
| ▪ Mahakali: Sharda | ▪ Kankai: Biring |
| ▪ Babai: Sarayu | |

Arun

- The river that originates at the highest altitude of the world is Arun (27690 feet)
- Largest tributary of Koshi River
- Arun is called Fungchu in Tibet
- Its total length is 513km (its 361km lies in Tibet and 152km lies in Nepal)

Facts and Figures on rivers of Nepal

- The person to highlight the fact that "83,000 Megawatt electricity can be produced from the rivers of Nepal is Dr. Hariman Shrestha.

- All the rivers of Nepal end in Ganga River in India. Ganga River finally ends in Indian Ocean.
- Rivers originating from Bhot peak (Tibet): Tamor, Arun, Tamakoshi, Sunkoshi, Trishuli, Karnali (6).
- The eastern most first class river that enters Nepal from Olangchungola pass (Taplejung) is Tamor.
- River of Nepal known as Uttar Ganga - Sano Bheri.
- River of Nepal known as Uttar Bahini – Karmonasha.
- The only river of Nepal that flows in opposite direction (South to North) is karmonasha. This river originates from Godawari of Lalitpur district and ends in Manohara River.
- The stream that ends in fewa lake is Harpan Khola and Marshi khola
- The river that flows in two different places with a same name is Rapti
- The famous religious spot "Ruru Kshetra" is located at the bank of Kali Gandaki River.
- The two rivers of Nepal listed in the world's top 10 most adventurous rivers for rafting are Sunkoshi and Karnali.
- Likhu River is also known as Barun/Baruwa.
- Chandra Canal of Saptari district flows from Triugya River.
- One pillar bridge, this name is imparted to a bridge constructed in Karnali river.
- Sunkoshi and Indrawati meet at a place called "Dolalghat".

Important Places located on the bank of different river

- | | |
|---|-----------------------------------|
| ▪ Kathmandu: Bagmati | ▪ Bhaktapur: Hanumante |
| ▪ Beshi Sahar (Lamjung): Marsyandi | ▪ Damauli (Tanahun): Madi |
| ▪ Pokhara (Kaski): Seti Gandaki | ▪ Kushma (Parbat): Modi |
| ▪ Panauti (Kavre): Padmawati | ▪ Kohalpur (Banke): Mahakali |
| ▪ Dhangadhi: Mohana | ▪ Mahendranagar-Mahakali |
| ▪ Tikapur (Kailali): Karnali | ▪ Jomsom (Mustang): Kaligandaki |
| ▪ Taulihawa (Kapilvastu): Banganga | ▪ Dunai (Dolpa): Thuli Bheri |
| ▪ Jumla: Tila | ▪ Jumla Khalanga: Sano Bheri |
| ▪ Khandbari (Sankhuwasabha): Arun | ▪ Biratnagar (Morang): Bakraha |
| ▪ Manthali (Ramechhap): Tamakoshi | ▪ Tumlingtar (Sankhuwasaba): Arun |
| ▪ Bidur (Nuwakot)-Trishuli | ▪ Sindhuli: Kamala Mai |
| ▪ Dhanusha: Hardinath | ▪ Butwal and Bhairahawa: Tinau |
| ▪ Dhunche (Rasuwa): Trishuli | ▪ Simikot (Humla): Humla Karnali |
| ▪ Chandragadhi, Kakadvitta (Jhapa): Mechi | ▪ Gaighat (Udayapur): Triyuga |
| ▪ Dipayal (Doti): Seti | ▪ Chame (Manang): Marsyangdi |
| ▪ Beni (Myagdi): Kaligandaki | |

Place of origin of some important rivers

- | | |
|---|---|
| ▪ Arun – Tibet | ▪ Trishuli - Gosainkunda / Langtang peak |
| ▪ Bishnumati - Sapan Tirtha | ▪ Likhu – Rolwaling mountain range |
| ▪ Bagmati – Baghdwar | ▪ Kaligandaki – Mustang lake |
| ▪ Budhi Gandaki - Ganesh Himal | ▪ Dudh Koshi - Dudh Kunda |
| ▪ Indrawati - Jugal Himal (Paanch Pokhari Daha) | ▪ Jhimruk – Goumukhi |
| ▪ Mahakali - Api Himal / Jaskar Himal | ▪ Marsyandi - Muktinath peak / Damodar peak |
| ▪ Thuli Bheri - Kanjirowa Himal | ▪ Tamor - Kumbhakarna peak |
| ▪ Sano Bheri - Dhaulagiri peak | ▪ Mechi - Lali Kharka |

- Mugu Karnali - Laddhag peak
- Humla Karnali - Tibet
- Banganga - Chure Hill (Arghakhanchi)
- Budhi Ganga - Jagdullaha Daha
- Kankai - Mahabharat Range
- Darundi - Buddha peak
- Seti Gandaki - Annapurna peak
- Kamala - Sindhuligadhi

Water supply

- 93.4 percent people have access to the basic water supply.
- 24.8 percent population has access to high or medium level quality water supply.
- 42,857 people have been connected to the basic water supply facility.

Electricity/Energy/Hydro energy in Nepal

- Rivers are the major source of hydro energy/hydroelectricity.
- The Roadmap for Energy Development has been implemented by declaring the **Energy and Water Resources Decade (2018-2028)/(2075-2085)** with the objective of contributing to fulfill the national aspirations of 'prosperous Nepal and happy Nepali' through the development of energy sector.

Electricity

- Total electricity connected to the national grid has increased by 30.9 percent to reach 2,205 megawatts until 2078 falgun. It was 1,685 megawatts as of Asar 2078.
- Out of the total generated electricity,
 - ❖ 2023 megawatt is from hydroelectricity,
 - ❖ 49.76 megawatt is from solar plant,
 - ❖ 53.4 megawatt is from thermal plant,
 - ❖ 74 megawatts is from renewable energy and
 - ❖ 6 megawatt as the by-product of the sugar mills
- Power generated by projects owned by Nepal Electricity Authority (NEA) is 581 megawatt and private sector owned hydropower project 1440 megawatt is connected with the national grid.
- Until 2078 falgun, the national transmission line (66 KV and above) has reached 4,795 circuit kilometers. The distribution line (33 KV and below) has reached 163,202 kilometers.
- The national transmission line (66 KV and above) had reached 4,573 circuit kilometers. The distribution line (33 KV and below) had reached 167,191 kilometers until last year (2077 falgun).
- As of falgun 2078 of the current fiscal year, the electricity imported from India is 757 gigawatt hours and the electricity exported to India is 183 gigawatt hours.
- In the last fiscal year, the electricity imported from India to Nepal was 2,826 gigawatt hours and the electricity exported to India was 44 gigawatt hours.
- Until falgun 2078, the electricity production capacity has remained 2,023 megawatt. It was 1,388 megawatt until asar 2078.
- In the current fiscal year 2078/79, 843 megawatt of hydroelectricity will be added from 18 power projects.
- In the next fiscal year 2079/80, 713 megawatt of hydroelectricity is expected to generate from different 43 power projects.

Access to electricity

- About 94 percent population has access to electricity until 2078 falgun with the increase in the electricity production, including the alternative energy.
- Per capita electricity consumption has reached 270-kilowatt hour.
- The access to electricity is highest 99.66 percent of the population in Madhesh Province and the lowest 43.87 percent in Karnali province in 2021/22

Province wise access to electricity (in percent)

Province	2019/20 (%)	2020/21 (%)	2021/22 (%)
1	86.3	82.43	96.95
2	87.25	99.1	99.66
Bagmati	95.83	94.44	95.91
Gandaki	87.48	92.79	95.68
Lumbini	89.07	91	94.91
Karnali	27.74	34.75	43.87
Sudur Paschim	67.33	64.69	71.07

Total Hydropower generation from different provinces (by mid-march of 2022)

Province	Electricity produced (Megawatt)	Province	Electricity produced (Megawatt)
1	327	Lumbini	31.0
2	21	Karnali	11.0
Bagmati	1016	Sudur Paschim	52.0
Gandaki	565	Total	2023

- Bagmati province stands first in terms of hydropower generation and Karnali Province stands at the last of the list of electricity production.

Status of Energy Consumption until falgun 2078

- As of mid-March of fiscal year 2021/22, total energy consumption has reached 5,772 gigawatt hours. Such consumption was 7,268 in the last fiscal year 2077/78.

Ratio of conventional, commercial and renewable energy consumption to total energy consumption (2077/78)

- Traditional (Conventional) energy consumption is 68.6%,
- Commercial energy consumption is 28.2% and
- Renewable energy consumption is 3.2%.

Annual average Electricity Consumption by different sectors of economy in 2078/79:

- Domestic sector: 41.0%
- Industrial sector: 38.0%
- Commercial Trade sector: 7.0%
- Other sectors: 10.0%
- A total of 5,572 gigawatt hours of electricity has been consumed by all sectors of the economy until falgun 2078

Alternative/Renewable Energy: As of mid-March of 2022,

- 6.9 percent of the total population has access to electricity from renewable energy sources. It was 3.2 percent in the last fiscal year.
- The percentage of population using clean renewable energy for various purposes has reached 36.0 percent in the fiscal year 2077/78.
- 398 kilowatts of electricity has been generated from micro and small hydropower under alternative energy in the current fiscal year 2078/79. Such production was 955 kilowatt in the last fiscal year 2077/78.
- 200 kilowatts of electricity has been generated from solar and wind energy until falgun of the current fiscal year. Such production was 284 kilowatt in the fiscal year 2077/78.
- 1,733 biogas plants have been adjusted in the current fiscal year 2078/79. Such adjustment was 3,975 in the last fiscal year.
- 11,956 Household solar panel systems have been adjusted in the current fiscal year 2078/79. Such adjustment was 12,567 in the last fiscal year.

Major points on Hydropower/Hydro Energy/Hydroelectricity

- The foreign currency obtained by exporting/selling hydroelectricity is called White Dollar.
- The largest hydropower project of the world is Three Georges, 22500 megawatt (China).

Farping Hydroelectric Project

- The first hydro power project of Nepal was established in 1968 B.S. (1911 A.D.).
- This hydropower was constructed under the technical aid of United Kingdom.
- Farping hydropower (Chandra Jyoti) was inaugurated on Jetha 9, 1968 B.S. After then Jetha 9 is observed as Energy day in Nepal.

Facts and figures on Hydropower Projects

- The largest hydropower project of Nepal currently in operation is Kaligandaki Hydropower, 144 Megawatt (Syangja).
- First Hydropower project constructed by the private sector in Nepal is Khimti, 60 Megawatt, Dolakha
- Hydroelectric project at the highest altitude is Thame, Solukhumbhu, 12,200 feet
- Largest producer of electricity in the world is China.
- Largest consumer of electricity in the world is United States of America.

Classification of Hydroelectric Projects

Micro Hydro power	1-100 Kilowatt
Small Hydro Power	1-25 Megawatt
Medium Hydro power	25-100 Megawatt
Large Hydro power	100-1000 Megawatt
Massive Hydro power	1000 Megawatt - above

Major Hydroelectric Projects of Nepal

S. No	Hydropower	Location	Capacity (MW)	Established date (A.D)	Donor agency
2	Farping	Kathmandu	0.5	1911	Nepal/UK
3	Kaligandaki	Syangja	144	2001	Nepal/ ADB
4	Mid- Marsyangdi	Lamjung	70	2009	Nepal/ Germany
5	Marsyangdi	Tanahun	69	1989	Germany/ World Bank
6	Khimti	Dolakha	60	2000	Private sector
7	Kulekhani- 1	Makwanpur	60	1982	World Bank
7	Bhotekoshi	Sindhupalchowk	36	2000	Nepal/Korea
8	Kulekhani-2	Makwanpur	32	1986	Japan
9	Chilime	Rasuwa	22	2003	Private sector
10	Trishuli	Nuwakot	21	1968	India
11	Gandak	Nawalaparasi	15	1979	India
12	Devighat	Nuwakot	14.1	1983	India
13	Modi	Parbat	14	2000	Nepal/ Korea
14	Jhimruk	Pyuthan	12.30	1994	Butwal power company
15	Sunkoshi	Sindhupalchowk	10	1973	China
16	Indrawati	Sindhupalchowk	7.50	2002	Private sector
17	Puwa Khola	Illam	6.20	1999	Nepal
18	Aandhi Khola	Syangja	5.10	1991	Butwal Power company
19	Sundarijal	Kathmandu	0.60	1936	Nepal

Completed Hydro Power Projects of Nepal

S.No	Hydro power	District	Generation Capacity (MW)
1	Syouri Bhumi	Nuwakot	0.02
2	Sobuwa Khola	Taplejung	0.09
3	Dhunge Jiri	Dolakha	0.6
4	Bishnu Priya Solar Farm	Nawalparasi	0.96
5	Chake khola	Ramechhap	2.8
6	Midim Khola	Lamjung	3.0
7	Sardi Khola	Kaski	4.0
8	Puwa Khola	Illam	4.0
9	Sabha Khola	Sankhuwasabha	4.0
10	Fawa Khola	Taplejung	5.0
11	Upper Mai 'C'	Illam	5.1
12	Super Mai	Illam	7.8
13	Rudi Khola	Kaski	8.8
14	Thapa Khola	Myagdi	13.6
15	Bagmati Small Hydropower	Lalitpur	22.0
16	Chamelia	Darchula	30.0
	Total		111.75

Source: Ministry of Energy, Water resources and Irrigation 2077

Hydropower Projects completed by mid-March Fiscal Year 2019/20

S.No	Hydropower	District	Production Capacity (KW)
1	Lower Hewa Khola	Panchthar	22100
2	Lower Chhote Khola	Gorkha	997
3	Upper Mardi	Kaski	7000
4	Iwa Khola	Taplejung	9900
5	Upper Naugarh Gad	Darchula	8000
6	Kabeli B1	Panchthar	25000
7	Juligad Small Hydropower Project	Bajhang	996
8	Padam Khola	Dailekh	4800
9	Rudi B	Lamjung, Kaski	6600
10	Ghalemndi	Myagdi	5000
11	Kulekhani III	Makwanpur	14000
	Total		104393

Source: Nepal Electricity Authority, 2020

Hydropower under construction to be completed in the FY 2076/77

S.No	Hydropower	District	Production Capacity (MW)
1	Upper Tamakoshi	Dolakha	456
2	Rasuwa Gadhhi	Rasuwa	111
3	Lower Solu	Solukhumbu	82
4	Sanchzen	Rasuwa	42.5
5	Mistri Khola	Myagdi	42
6	Khani Khola 1	Dolakha	40
7	Upper Balefi A	Sindhupalchowk	36

8	Khani khola	Dolakha	30
9	Dordi Khola	Lamjung	27
10	Upper Dordi A	Lamjung	25
11	Solar Energy	Nuwakot	25
12	Singati Khola	Dolakha	16
13	Upper Sanchzen	Rasuwa	14.8
14	Upper Khimti	Ramechhap	12
15	Lower Khare	Dolakha	11
16	Jun Beshi	Solukhumbhu	5.2
17	Rukum Gaand	Rukum	5
18	Lower Tadi	Nuwakot	5
19	Tadi Khola	Nuwakot	5
20	Pikhuwa Khola	Bhojpur	5
21	Khoranga Khola	Terhathum	4.8
22	Rawa Khola	Khotang	3
23	Tang Chhahara Small Hydro	Mustang	2.4
24	Upper Parajuli Khola -1	Dailekh	2.2
	Total		1007.8

Major Hydro projects under construction

- Paschim Seti (750 Megawatt)
- Upper Modi A (42 Megawatt)
- Budhi Gandaki (600 Megawatt)-Dhading and Gorkha
- Pancheswori (6480 Megawatt)
- Upper Trishuli I: 216 Megawatt

Other hydroprojects and their locations

- Gorkhe, Kabeli: Illam
- Nalsingh Gandh: Jajarkot
- Rupalghat: Dadeldhura
- Jhupra: Surkhet
- Namarjun Modi (12 Megawatt): Kaski
- Chatara: Sunsari
- Arughat: Gorkha
- Syarpu Daha: Rukum
- Tatopani: Myagdi

Decade of Energy and Water Resources

To promote the slogan of "Prosperous Nepal and Happy Nepali"

- 2075-2085 is observed as "the Decade of Energy and Water resource"

Lakes, ponds, kunda, Daha:

- Lakes and ponds are also important sources of water.
- There are many lakes and ponds in Nepal.

Rara Lake: Largest Lake of Nepal located in Mugu District

- Length: 8km (water spread in 5.2km), width/breadth: 1.5km, depth: 167 meter
- This lake is situated at a height of 2990 meters above the sea level.
- This lake was named Mahendra Lake in 2020 Chaitra 13.
- The stream that flows out of this lake is Khatyad Khola.
- This is the lake at the highest altitude of the world where fishes are found.

Shey - Phoksundo Lake: Deepest Lake of Nepal located in Dolpa District

- Length: 4.8Km, Width/breadth: 1.6km and depth: 650 meter
- This lake is situated at a height of 3613 meters above the sea level.
- This lake originates from Kanjirowa peak (6883meter), Dolpa.

- Also known as Rigma lake, Nirjib lake, Y/catapult shaped lake (has three corners).
- Shuligand water falls (167m) originates from this lake at a height of 3613 meter.
- Shuligand is the water falls that originates at the highest altitude of Nepal.

Tilicho Lake: Lake situated at the highest altitude of the world

- It is situated at a height of 4919 meter (Manang).
- Length: 4km, Width: 1.2km, Depth: 200meter.
- This lake is called Tilijo or Tirichho in local language.

Largest Man Made lake/water reservior:

- Indra Sarovar situated in Makwanpur district.
- This reservior is made to store water for Kulekhani Hydropower Project.

Largest Glacier Lake:

- Chho-rolpa Lake: Dolakha

Major lakes (Taal) and ponds (Pokhari) of Nepal and their location:

- Myagdi : Bhale Basne Taal
- Kapilvastu : Jagadisapur lake, Jakhira Lake
- Dhanusha : Ganga Sagar
- Dolpa : Jagadullaha, Phoksundo lake,
- Udayapur : Rautaha, Suke, Jata Pokhari
- Kanchanpur : Jhilamila, Banda, Rani (lake)
- Gorkha: Kalchhuman lake, Rukche lake
- Rupandehi : Bhoot Kuwa, Gaidahawa Lake
- Sunsari: Chimdi Barju lake, Taltalaiya
- Dang: Barhakune Lake, Bhote Daha, Charingre Daha, Kichkanni Daha.
- Kaski - Phewa, Rara, Beganas, Dipang, Maidi, Budhwa, Khyaste, Syarkhu (lake)
- Baglung : Bhuinfutta Lake, Rudra lake, Jalpa Pokhari
- Chitwan : 20 Hajari, 16 Hajari, Tamorghaila, Gadhuwa, Kasara, Mujura, Nanda vauju, Baikuntha, Bandan(lakes), Devi lake, Munda lake, Tamar lake, Tami lake
- Sindhupalchowk : Paanch pokhari (Jata, Bhoot, Kala, Bhale, Rani) Tara khasne pokhari, Sipa pokhari, Narayan Pokhari
- Taplejung : Manas, Sijme lake, Jaljale, Suke, Nurak, Siwa, Kali, Rale, Iwa, Tin, Pancha Pokhari
- Kathmandu : Pani Phokari, Kamal Pokhari, Rani Pokhari, Nag Pokhari, Gahana Pokhari
- Kapilvastu: Jagadishpur lake (transit home for the birds from Siberia), Jakhira Lake
- Sankhuwasava : Sabha, Siddha, Matsya, Gosa, Banduke, Kata, Jumlyaha Pokhari
- Parsa: Ghadiarwa lake, Chhapkaiya lake, Baighimai Pokhari, Banjari lake
- Arghakhanchi: Gauchaur, Kamal Pokhari
- Bajhang: Chhubhal lake, Surma Sarowar, Rakshya lake, Khapar, Nilkhathi Lake
- Bhaktapur: Siddha Pokhari, Bhaju Pokhari, Kamal Pokhari
- Dolpa: Shey Phoksundo, Jagadullaha, Rigma lake, Putha lake, Pungma lake
- Terhathum: Chingling, Tiche, Siddamarga, Chitre, Lam Pokhari
- Surkhet: Jajura, Baraha, Bulbule lake
- Solukhumbu: Dudh, Goke, Imja, Gokyo lake, Jalim, Namche, Salpa Pokhari
- Panchthar: Jore, Suke, Namdu, Ratamate, Baraha Pokhari
- Achham: Ramroshan/Jigale lake
- Mustang: Dhumba lake
- Rukum: Syarpu lake
- **Bharat lake: Bagmati municipality, Sarlahi (man made lake)**
- Lalitpur: Nhu Pokhari
- Illam : Mai Pokhari
- Bajhang : Surma Sarovar
- Doti: Khaptad Lake
- Kailali: Ghoda Ghodi lake
- Surkhet: Bulbule lake
- Palpa: Satyawati, Suke Taal, Nandan Lake
- Dhading: Ganga Jamuna Kunda
- Sindhuli: Gajuli Daha

Kunda

- Damodar Kunda: Mustang
- Dudh Kunda: Solukhumbhu
- Ganga Jamuna Kunda: Dhading
- Gosainkunda, Bhairab Kunda, Surya kunda: Rasuwa
- Shanti Kunda: Kaski
- Pap Kunda: Mugu
- Narad Kunda: Gorkha
- Gyan Kunda: Dhanusha

On the auspicious occasion of Janai Purnima, Hindu pilgrims from Nepal and India gather in Gosainkunda Lake to take holy bath. Janaipurnima Festival is celebrated as Fair in Gosainkunda.

Daha (pond/form of lake)

- Kathmandu: Tau Daha, Katuwal Daha, Indra Daha
- Bara: Kamini Daha
- Jumla: Giri Daha, Sekh Daha
- Kanchanpur: Betkote Daha
- Bajhang: Khaptad Daha
- Rupandehi: Gajedi Daha
- Jhapa: Gaura Daha, Chilagadh Daha
- Rupandehi: Gajedi Daha
- Doti: Simri Khani Daha, Sun Daha, Khaptad Daha
- Arghakhanchi: EEpi Daha, Khandaha, Jhirradaha
- Mugu: Rin Mokshya Daha, Chhayanath Daha
- Kalikot: Dhauli Daha, Jogini Daha, Bista, Sankhu Daha
- Mahottari: Sishuwa Khataiya Daha, Gudke Daha
- Rukum: Syarpu Daha, Rukmini Daha, Rukumkot Daha
- Achham: Ram Roshan Daha, Jigale Daha, Taule Daha
- Humla: Jadya Daha, Selim Daha, Lade Daha, Nilapani
- Dang: Bhote Daha, Charingre Daha, Kichkanni Daha, Syanglang Daha
- Lalitpur: Nag Daha
- Salyan: Kuvinde Daha, Kachhuwa Daha
- Udayapur: Jogi Daha
- Surkhet: Simri Daha, Murali Daha
- Sindhuli: Gajuli Daha
- Saptari: Ghoda Daha
- Dolpa: Pritha Daha
- Gorkha: Nag Daha

Facts and Figures

- The lake that changes its appearance throughout the day - Jhilamila Lake, Kanchanpur.
- Ghodaghodi Lake is also known as Kapil Lake. Ghodaghodi Lake has a shape of palm of hand.
- Pachvaiya Hill separates Beganas and Rupa lake of Kaski District.
- Fewa Lake is formed by the water from Harpan and Mardi Stream (khola)
- Tal Barahi Temple lies in the middle of Fewa Lake. So Fewa Lake is also called Barahalake.
- Fewa lake is commonly known as Baidam lake among the local population.
- Histroical Rani Pokhari was built by Pratap Malla to console his queen Anantapriya, who was highly depressed by the death of her son chakrabartendra.

Glaciers

- The largest glacier of Nepal is Langtang glacier, Rasuwa (18km).
- The longest glacier of Nepal is Khumbu glacier, Solukhumbhu (32km).
- Currently The Ngozumpa glacier that lies below the sixth highest mountain in the world Cho Oyu, at 36 km has been listed as the longest glacier in the Himalayas. (Source: ICIMOD)
- **Note:** whenever the longest glacier is asked the first priority goes to Khumbu, but if there is no khumbu in option, go for Ngozumpa as the longest glacier of Nepal.
- Glacier at the highest altitude of the world is Khumbu glacier. It is at a height of 7600 meter.
- Chho rolpa snow lake, Dolakha is the largest glacier lake of Nepal.
- Fungi Glacier lies in Gorkha and Chabuk glacier lies in Taplejung.
- Amodeblem Glacier lies in Gorkha.

Water falls:

- The largest waterfall of Nepal is Hyatung falls (Terhathum), 365 meters.
- Hyatung water falls is also called "Hingwa".
- The longest waterfalls of Nepal is Pachal water falls (Kalikot), 381 meters
- The water falls originating at the highest altitude is Shuligand falls (Dolpa), nearly 167 meters long. This water falls originates at a height of 3613 meters. Shuligand water falls is called Chhalchhaa and Pauli water falls.

Major water falls of Nepal

- | | |
|---|--|
| ▪ Namaste (Simsuwa): Dhankuta | ▪ David Falls, Khudi: Kaski |
| ▪ Tiptala, Mewakhola: Taplejung | ▪ Pokali, Ratamata, Sepli: Okhaldhunga |
| ▪ Chirkhuwa: Bhojpur | ▪ Purandhara: Dang |
| ▪ Diprung: Khotang | ▪ Khuwa, Odari: Udayapur |
| ▪ Fung fung: Nuwakot | ▪ Jorpani: Doti |
| ▪ Paanch Pokhari: Ramechhap | ▪ Suligaand : Dolpa |
| ▪ Illam Danda: Taplejung | ▪ Ganga Jamuna: Dhading |
| ▪ Teel Khuwa, Sheet Khola: Arghakhanchi | ▪ Ghyamche: Mugu |
| ▪ Seti river falls: Bajhang | ▪ Baikuntha: Palpa |
| ▪ Chharchhare: Palpa | ▪ Sunachhahari: Rolpa |
| ▪ Bhairab Kunda, Gumchha : Sindhupalchowk | ▪ Todke: Illam |
| ▪ Todke, Simle, Betini, Asle: Illam | ▪ Boksi: Myagdi |
| ▪ Chepe river water falls: Manang | ▪ Ghyamche: Mugu |
| ▪ Gaikhure: Lamjung | ▪ Panch Pokhari Jharana: Ramechhap |
| ▪ Tindhare: Kavrepalanchowk | ▪ Chirkhuwa: Bhojpur |

Soil

- Soil is the loose surface material that covers most land.
- Scientific discipline that studies all aspects of soils, including their physical and chemical properties is called Pedology.
- Edaphology is the science or study of soil, especially with respect to plant growth.
- Soil conservation program in Nepal was launched in 2002 B.S.
- World Soil day is observed on December 5 every year.
- World Soil day was first observed in 2014 A.D.
- The theme of 2021 world soil day is "Halt Soil salinization, boost soil production).

Classification of Soil in Nepal**Alluvial Soil:**

- A fine grained fertile soil deposited by water flowing over flood plains or in river beds.
- This type of soil is formed by the deposition of weeds, creepers and grass in the sand.
- This type of clay is mainly found in Eastern Terai region.
- The best soil for the production of paddy crops, Jute, sugarcane, oil products.

Sandy Boulder Soil

- This type of soil is less fertile or has no fertility.
- This type of soil does not absorb water.
- This type of soil is suitable for the growth of trees like Sal, Sissoo etc.

Red Brown soil

- This type of soil is favorable for fruit farming and cash crops.
- Tea, cardamom, coffee, potato, maize and millet are the major crops grown in this soil.

Lacustrine Soil

- Soil formed on or from lacustrine deposits, i.e. material deposited in lake water and later exposed.
- This type of soil has more depth and permits very fine particles (fine sand, silt and clay).
- This type of soil is appropriate for Paddy, Wheat and vegetable growth.
- Agricultural products can be grown in such soil even in less rainfall.
- This type of soil is found in Kathmandu valley.

Glacial Soil

- Soil composed of boulder clays, moraines etc., which are formed by the action of ice and are found at the base of mountains.
- This type of soil is less fertile.
- Barley, Buck wheat, potato, millet, maize are grown in this type of soil.

Forest:

- There is a famous quote "Hariyo Ban Nepalko Dhan" i.e. is Green forest is the wealth of Nepal.
- Forest is a very important natural resource of Nepal.
- Forest provides wood/furniture, fuel, paper, herbs, matches etc.
- It influences climates, causes rain, stops flood, soil erosion and landslide.
- Forest regulates the temperature of the surrounding areas.
- 30.8% of the total area of Earth is surrounded by forest.
- There are 422 trees per hectare in the world.
- Every individual equals to 0.6 hectare of forest in the world.
- Minimum area of forest required for healthier living is 43% of the total area of land.

Of the total land area of Nepal

- **Forest area covers 44.8 percent of which**
 - ❖ Dense forests cover 40.4 percent and
 - ❖ Bushes and sapling cover 4.4 percent.
 - ❖ The forest area covers 5.962 million hectares of land and other bush and saplings area cover 648,000 hectares.
 - ❖ per capita forests of Nepal is 0.2 hectare and 111 trees
- **In terms of topography of Nepal,**
 - ❖ the largest forest area is in the mid-hills (37.8%) and
 - ❖ The smallest forest area is in the Terai (6.9%).
- **Of the total forest area,**
 - ❖ The largest part of forest lies in Karnali province and
 - ❖ The smallest part is in Province 2.
- **On the basis of total land area of province**
 - ❖ the highest part of forest lies in Sudurpaschim province and
 - ❖ the lowest lies in Province No. 2

World Context

- Forests cover 30.8 percent of the total area of the world.
- The per capita forest area is 0.6 hectare and 422 trees.
- Of the total land area, 22.1 percent is covered by forests in neighboring China.
- Among the South Asian nations, Bhutan has the highest, 72.5 percent forest areas whereas Pakistan has the lowest 1.8 percent.

Source: Forest Department/forest research and survey development (Fiscal year 2021/22)

National Parks, Conservation Areas, Wildlife Reserve and Hunting Wildlife Reserve, Buffer areas:

Total area occupied by National Park, Conservation Area, Wildlife Reserve and Buffer areas

- Area occupied by National Park: 11, 806.02 Sq.km
- Area occupied by Conservation Areas: 15, 434.65 Sq.km
- Area occupied by Wildlife reserve: 175 Sq.Km
- Area occupied by Hunting Wildlife Reserve: 1325 Sq.km
- Area occupied by 13 buffer areas: 5, 687. 78 Sq.km
- Total area occupied: 34428.45 Sq.km (23.4 % of the total area of Nepal).

National Parks

National Parks	Area (Sq.Km)	Established Date (B.S)	Districts covered
Chitwan National Park	952.63	2030	Chitwan, Makwanpur, Parsa, Nawalparasi East (Nawalpur)
Langtang National Park	1710	2032	Rasuwa, Nuwakot, Sindhupalchowk
Sagarmatha Natinal park	1148	2032	Solukhumbu
Rara National Park	106	2032	Mugu and Jumla
Bardiya National Park	968	2032	Bardiya
She- Phoksundo National park	3555	2040	Dolpa and Mugu
Khaptad National Park	225	2042	Bajhang, Bajura, Achham, Doti
Makalu- Barun National Park	1500	2049	Sankhuwasaba and Solukhumbu
Shivapuri- Nagarjuna National Park	159	2058	Kathmandu, Nuwakot, Sindhupalchowk, Dhading
Banke National Park	550	2067/3/28	Banke, Salyan, Dang
Shuklaphanta National Park	305	2031	Kanchanpur
Parsa National Park	627.39	2040	Parsa, Makwanpur, Chitwan

National Parks	Famous for	Other Features
Chitwan National Park	One horned Rhino, Tiger, Ghadiyal, Hilakewa bird, Striped Hyena, Chital, Hog deer, Barking deer, Langur, Rhesus monkey	First National park of Nepal. 2 nd National park included in world heritage site in 1984 A.D. National park visited by most number of tourists.
Langtang National Park	Red Panda, Musk deer, Ghoral, Wild dog Reus monkey, Yarsagumba, paanchaule	2 nd largest national park Beden powell scout peak, Gosainkunda, Bon Po Gumba, Langtang glacier lie inside this park
Sagarmatha Natinal park		Listed in world heritage site in 1979 A.D. National park at the highest altitude. Famous monasteries : Tengboche, Pangboche, Thame lie in this national park
Rara National Park	Musk deer, wild boar, Himalayan Thar, Kalij	Rara lake lies in this park, the lake in highest altitude where fishes are found (Snow trout fish)

Bardiya National Park	Rhino, Tiger, Swamp deer, Black buck, Ghadial, Gangetic Dolphin	
She- Phoksundo National park	Snow leopard, langur, Tibetan hare, Serow, Wolf, Jackal, Himalayan Blue Sheep (Naur), Himayan Thar, Himalayan Black bear, Himalayan weasel, yellow throated marten	Largest national park She phoksundo lie in this national park
Khaptad National Park	Leopard, Himalayan Black bear, wild dog, Musk deer	
Makalu- Barun National Park	Snow leopard, Clouded leopard, jungle cat, Himalayan wolf, red fox, black beer, deer, wild boar, spotted linsong, Weasel, Marmot.	Two rare species of birds, Rein Barbler and Olive Ground Barbler are found here There are 12 villages in this national park.
Shivapuri- Nagarjuna National Park	various species of snakes, lizard, skink toad, frog, eagle owl, slender billed scimitar-bablar, white gorgeted flycatehe dove, Golden throated barbet.	Established as shivapuri wildlife reserve in 1972 AD (2029 BS) Converted to national park in 2058 BS (2002 AD) On 2065 Poush 23, Nagarjun forest of Area 15 sq.km was unified to form shivapuri-Nagarjun National park. Bagdwar, Sapantirtha, Nagi Gumba, Sundrijal lies inside this park.
Banke National Park	Tiger	
Shuklaphanta National Park	Swampdeer, Chittal, arasinghe wild elephant, tiger, leopard, wild bear hispid hare, Bengal florican, marshmugger, Indian python, Cobra	Historically important Rani lake lies here The largest grass land of Nepal lies here which is 54 sq.km
Parsa National Park	wild elephant king cobra, krait,python, sambar deer, chital deer, hog deer, barking deer, laguna, hyena, palm civet	

Facts and Figures

- The first National Park of Nepal is Chitwan National Park
- Parsa Wildlife Reserve has been transformed to Parsa National Park in 2074 Jetha 20. This is the last National Park of Nepal
- Karnali wildlife Reserve established in 2032 B.S. have been transformed into Bardiya National Park in 2045 Mangsir 20
- Bardiya National Park is the largest National Park of Tarai Region
- She-Phoksundo is the largest National Park of Nepal (3555Sq.km)
- The smallest National Park of Nepal is Rara National Park (106Sq.km)
- Area of Parsa National Park has been expanded to 627.37 Sq.km from 499 Sq.km in 2072 Ashadh 15
- The current area (952.63Sq.km) of Chitwan National Park was expanded in 2073 Kartik 1 from 932 Sq.km

- The largest grassland of Nepal lies in Shuklaphanta National Park which is 54 Sq.km
- Sagarmatha and Chitwan National Park have been listed in the World Heritage site in 1979 A.D. and 1984 A.D. respectively

Conservation Areas

S.No	Conservation areas	Established date (B.S)	Area (Sq.km)	Districts covered
1	Annapurna	2049	7629	Kaski, Mustang, Manang, Lamjung, Myagdi
2	Kanchanjunga	2054	2035	Taplejung
3	Manaslu	2055	1663	Gorkha and Manang
4	Black Buck	2065/11/23	16.95	Bardiya (Khairapur)
5	Gaurishankar	2066/09/26	2179	Dolakha, Ramechhap and Sindhupalchowk
6	Apinampa	2067/03/28	1903	Darchula

Facts and figure on conservation areas

- Total area occupied by conservation areas: 15434.65 Sq.Km
- Annapurna is the second conservation area of the world, established on the occasion of 25th Jubilee of World Wildlife Fund (WWF). Muktinath, Damodar Kunda and Lomanthaka lie in this conservation area
- The frogs that are found only in Nepal are Paha Rostandi and Paha Blanfodi. These species of frogs are found in Kanichenjunga conservation area.
- Annapurna conservation is mainly famous for the conservation of Red panda (Habre).

Current Affairs

Godawari-Fulchowki area declared the 7th conservation area of Nepal

- Godawari-fulchowki area of Lalitpur and Kavrepalanchowk has been listed as the last conservation area of Nepal.
- This was listed as the last/7th conservation area of Nepal on 2079 Asar 02.

Wildlife Reserve

- The only wildlife reserve of Nepal is Koshi Tappu Wildlife Reserve
- Shuklaphanta and Parsa Wildlife Reserves have been upgraded to National Park

Koshi Tappu

- Established in 2032 B.S. with an area of 175 Sq.km
- Districts Covered: Sunsari and Saptari
- Objective: preservation of Buwalis Arna (Wild Buffalo)
- Famous for : king of Ducks, Chakhewa, Pochard, spotted deer, wildbear and blue bull
- Largest wet land area in Nepal lies in Koshi Tappu
- It was listed in Ramsar site in 1987 A.D.

Hunting Wildlife Reserve

- The only Hunting wildlife reserve of Nepal is Dhorpatan Hunting wildlife reserve
- The 2nd Hunting Wildlife reserve is being set up in Kushe Rural Municipal in Jajarkot. **The name of the reserve will be "Kushe Patan Hunting Wildlife Reserve".** Kushe Patan is at a height of 4388 meter and is linked to Jumla district

Dhorpatan Hunting Wildlife Reserve

- Established in 2044 B.S. with an area of 1325 Sq.km
- Districts covered: Baglung, Myagdi and Rukum
- The religious spot Dhor Baraha lies in this reserve
- Famous for blue sheep, ghoral, serow, barking deer, wildboar, rhesus macaque, langur monkey, mouse hare, wolf, pheasants and partridges, woodpeckers.

Wetlands

- Wetlands of Nepal Listed in Ramsar Site: 10
- The first wetland listed in world Ramsar site is Koshi Tappu wildlife reserve in 1987 December 17.
- Nine different lakes of kaski district considering single area were listed in 2016 February 02 as the last wetland site to be enlisted in world Ramsar site.

S.No	Wetlands	Date of enlistment in Ramsar site	Area (Hectar)
1	Koshi Tappu Wildlife reserve	1987 December 17	17,500
2	Twenty thousand lake (Bis Hajari Tal)	2003 August 13	3,200
3	Jagadishpur lake	2003 August 13	225
4	Ghoda Ghodi lake	2003 August 13	2563
5	Sagarmatha National Park (Gokyo Area)	2007 September 23	7770
6	Shey Phoksundo national Park	2007 September 23	494
7	Rara National Park	2007 September 23	1583
8	Gosainkunda, Rasuwa	2007 September 23	1030
9	Maipokhari, Illam	2009 October 28	90
10	Phewa Lake, Rupa lake, Beganas Lake, Dipang, Maidi, Khyaste, Neureni, Guaidee, Kamal pokhari	2016 February 2	26, 106

Protected Animals of Nepal

Mammals (Standhari): 26

Zoological Name	Common Name	Zoological Name	Common Name
Manis pentadactyla	Pangolin (salak)	Macaca assamensis	Asamese Monkey (Asami Rato Bandar)
Canis lupus	Grey Wolf (Khairo Bwanso)	Ursus arctos	Brown Bear (Khairo Bhalu)
Ailurus fulgens	Red Panda (Rato Panda)	Prionodon pardicolor	Spotted Linsang (pate Silu)
Hyaena hyaena	Striped Hyaena (Dhashre Hundar)	Prionailurus bengalensi	Leopard cat (Chari Bagh)
Pardofelis nebulosa	Clouded leopard (Dhwanse Chituwa)	Felis lynx	Lynx (Linx)
Panthera tigris	Royal Bengal Tiger (Bagh)	Panthera uncia	Snow leopard (Hiun Chituwa)
Platanista gangetica	Gangetic Dolphin (Shonsh)	Elephas maximus	Wild Elephant
Rhinoceros unicornis	One horned Rhinoceros	Moschus Chsyogaster	Musk Deer
Cervus duvauceli	Swamp deer	Bos gaurus	Gaur
Bos grunniens	Wild Yak	Bubalus arnee	Wild water Buffalo
Tetracerus quadricornis	Four horned Antelope	Antelope cervicarpa	Black Buck (Krishnasar)
Pantholops hodgsoni	Tibetan antelope (Chiru)	Ovis ammon	Great Tibetan Sheep (Nayan)
Caprolagus hispidus	Hispid Hare	Sus salvanicus	Pygmy Hog

Birds: 9

Zoological name	Common name	Zoological name	Common name
<i>Ciconia nigra</i>	Black Stork	<i>Ciconia ciconia</i>	White Stork
<i>Lophophorous impejanus</i>	Impeyan Pheasant	<i>Tragopan satyra</i>	Crimson horned pheasant (Danfe)
<i>Catreus Wallichii</i>	Cheer Pheasant	<i>Houbaropsis bengalensis</i>	Bengal floricon (Khar Mayur)
<i>Sypheotides indica</i>	Lesser Floricon	<i>Grus antigone</i>	Sarus crane
<i>Buceros bicornis</i>	Great pied Hornbill		

Reptiles: 3

Zoological Name	Common name	Zoological Name	Common Name
<i>Gavialis gangeticus</i>	Gharial	<i>Varanus flavescens</i>	Golden monitor lizard
<i>Python molurus</i>	Asiatic Rock Python		

Flora – 18

- Acacia Catechu (khayar)
- Silk cotton tree(simal)
- Rosewood(satisal)
- Orchid (paanchaunle)
- Gentian (kutki)
- Serpentine (sarpagandha)
- Himalayan Yem (Launth salla)
- Valerian(Sugandhawala)
- Walnut(Okhar)
- Cinnamomum glaucenscens (sugandhakokila)
- Fir (Tulsi patra)
- Cordyceps Sinensis (Yarsagumba)
- Walnut (Okhar)
- Golden champak(chaanp)
- Indian kino tree (Bijayasal)
- Common sal (saal)
- Spike ward (Jatamasi)
- Lichens(Jhyau)

Metals and Ores

- The first extracted metal is Copper.
- The first currency of Nepal "Manank" was made of copper.

Places famous for different Natural mines and Ores

- Places famous for different Natural Ores
- Glass: Ganesh Himal, Baitadi, Baglung
- Himalayan Black salt: Rasuwa, Syafrubeshi
- Soda: Salyan, Doti
- Pyrite: Syanja, Jhapa
- Turmalin: Sankhuwasava
- Haematite: Nuwakot, Tansen and Phulchowki
- Iron: Phulchoki (Lalitpur), Thoshe (Ramechhap), Labdhikhola (Tanahu), Jirbang (Chitwan), and Dhuwakot (Parbat), Bhainse (Makwanpur), Dhaubadi Area (Nawalparasi East)
- Copper: Buddhakhola(Bandipur), Okhaldhunga, Dhankuta, Myagdi
- Gold: Kali Gandaki, Budhi Gandaki, Bank of Sunkoshi river
- Marble: Godawari, different areas of Mahabharat range
- Limestone: Bhainse Dovan(Hetauda), Godawari, Palpa, Chovar, Jogimara
- Magnesite: Kharidhunga (Dolakha), Kyampughat(Udayapur)
- Cobalt: Gulmi, Arghakhanchi, palpa, Dhankuta
- Petrol: Nepalgunj, Koilabash, Dhangadhi, Dailekh, Mustang, Pyuthan, Dang
- Methane Gas: Kathmandu valley (Kathmandu, lalitpur, Bhaktapur)
- Coal: Dang, Chautara, chatara (Sunsari) Chitwan, Kathmandu valley
- Silver: Fulchowki, Chisapanigadhi, Baglung
- Graphite: Palpa, Sindhupalchowk
- Salt: Thak khola, Mustang
- Platinum: Palpa (Ridi)
- Gypsum: Dang, Salyan
- Nickel: Khokling, Nampa, Tunghang

- Uranium: Bank of Chameliya river Darchula, Mustang, Makwanpur
- Sulphur (Gandhak): Baraha Kshetra (Sunsari), Gosainkunda (Rasuwa)
- Cobalt: Dhankuta, Palpa, Gulmi, Arghakhanchi
- Slate: Bandipur, Bagmati, Lumbini, Gandaki, Janakpur
- Tin (Sn): Meddi and Genera (Dadeldhura) and Mandhukhola area (Makawanpur)
- Bismuth: Dadeldhura and Baraghare and Mandukhola area in Makawanpur district.
- Cinabar, Chief ore of mercury: Tirche Pani/Taruka
- Berllium: Manang, Kathmandu, Nuwakot, Rasuwa, Phakuwa Hyakule, Ilam and Taplejung districts